

Predicting Growth and Remodeling of Cardiovascular Tissue

change in mass

change in microstructure

Martin R. Pfaller

Assistant Professor

Department of Biomedical Engineering, Yale University

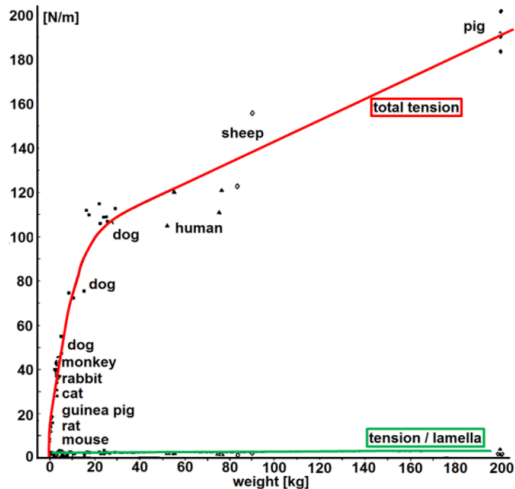
biomechanics.yale.edu

Student Version

Yale

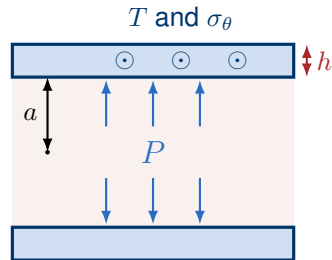
A Brief Introduction to Mechanobiology of Blood Vessels

Vascular tissue has a preferred mechanical state



Tension T varies by a **factor of 26**

$$T = P a$$

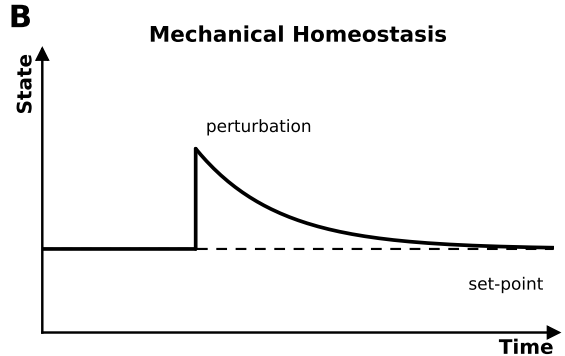
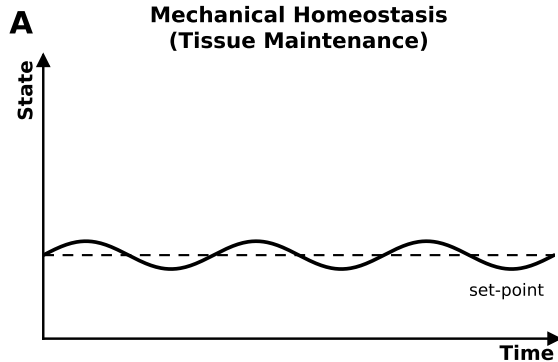


Tension / lamella varies by a **factor of 2.8**

$$\frac{T}{h} = \frac{P a}{h} = \sigma_\theta$$

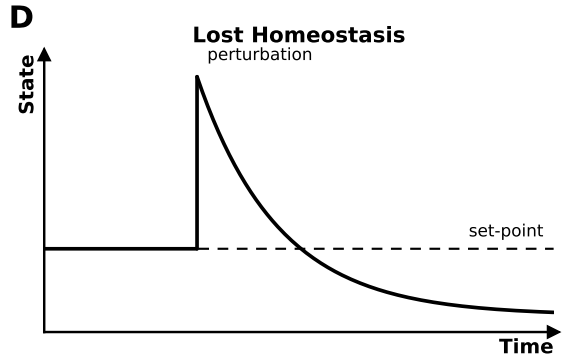
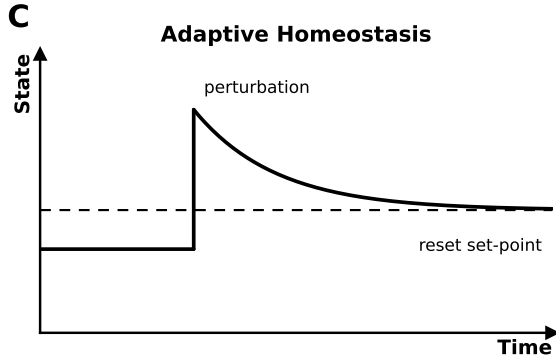
Mechanical Homeostasis

The tissue holds a mechanical **set-point**: small fluctuations are tolerated (**A**), and a perturbation relaxes back to it (**B**).

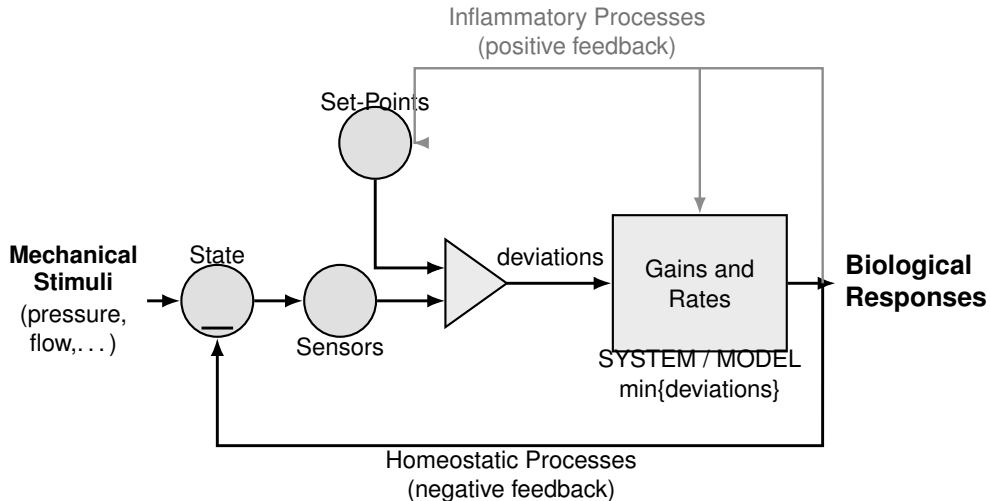


(Mal)adaptation

The set-point itself can shift: a perturbation may settle at a **reset** set-point (**C**), or, when regulation fails, the state runs away (**D**).

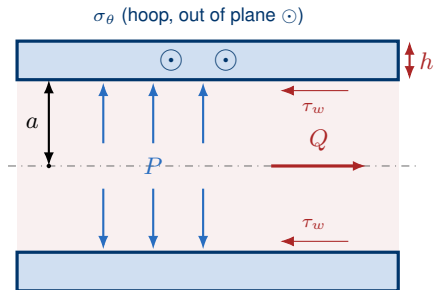


Mechanobiological Regulation



An Idealized Blood Vessel

A vessel is a **pressurised, perfused tube**. Two loads act on the wall: **blood pressure** P pushes it outward, and **blood flow** Q drags its inner surface. Cells sense both.



a inner radius, h wall thickness

P lumen pressure $\rightarrow$ **circumferential stress** σ_θ

Q volumetric flow $\rightarrow$ **wall shear stress** τ_w

Two Loads, Two Formulae, Two Adaptations

Pressure P sets the **hoop stress**; flow Q sets the **wall shear stress**. Each has a set-point; rearranging shows the geometry that restores it.

Pressure (Laplace)



Flow (Poiseuille)



✂ Your turn: Pressure adaptation

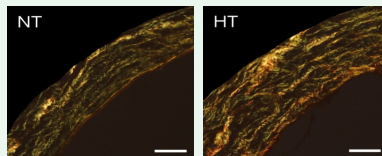
Question. Pressure **doubles**. How does the wall thickness change? $\sigma_\theta = \frac{Pa}{h}$

✓ Answer: Pressure adaptation

$P'/h' = P/h$ with $P' = 2P$ gives

(3)

Wall thickens *in proportion* to pressure.



$P \uparrow$

✂ Your turn: Flow adaptation

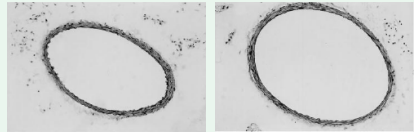
Question. Flow rises **8-fold**. How does the radius change? $\tau_w = \frac{4\mu Q}{\pi a^3}$

✓ Answer: Flow adaptation

$$(a'/a)^3 = Q'/Q = 8, \text{ so}$$

(4)

Reading. Radius grows as the *cube root* of flow.



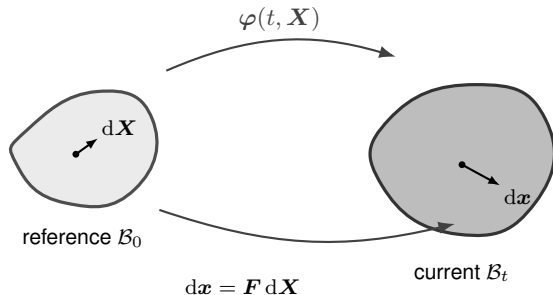
$Q \uparrow$

Finite-Strain Fundamentals

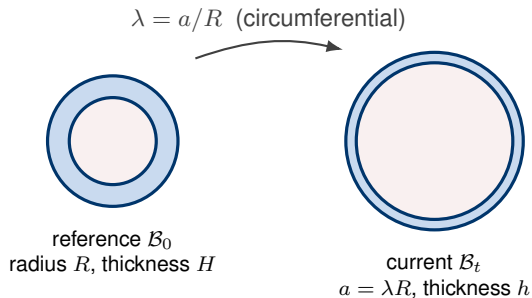
Reference and Current Configuration

A body is described by a **reference** $\mathcal{B}_0$ (points $\mathbf{X}$) and a **current** $\mathcal{B}_t$ (points $\mathbf{x}$) configuration, related by the motion $\mathbf{x} = \varphi(t, \mathbf{X})$.

3D 3D geometric model



0D 0D reduced model



The Deformation Gradient

The **deformation gradient** maps reference line elements to current ones.

3D a 3×3 tensor



(5)

J is the volume ratio, **incompressibility** means $J = 1$.

0D a scalar

$$\mathbf{F} \longrightarrow \lambda = \frac{\text{current length}}{\text{reference length}}, \quad \lambda = 1 \text{ undeformed.}$$

Strain: A Rotation-Free Measure of Stretch

Removing rigid rotation gives the **right Cauchy–Green** tensor and the **Green–Lagrange strain**.

3D symmetric 3×3 tensors



(6)

0D scalars

$$C = \lambda^2, \quad E = \frac{1}{2}(\lambda^2 - 1).$$

Stress: "Force per Area"

Cauchy stress: *current* force per unit *current* area.

3D spatial symmetric 3×3 tensor (from 2nd Piola–Kirchhoff stress S)

$$\boldsymbol{\sigma} = \frac{1}{J} \mathbf{F} \mathbf{S} \mathbf{F}^T, \quad \mathbf{S} = 2 \frac{\partial W}{\partial \mathbf{C}}, \quad W \text{ strain energy density function.}$$

0D scalar reduction ($C = \lambda^2$, $J = 1$) — the σ_θ in Laplace (1)

$$\sigma = \frac{1}{J} \lambda S \lambda = \lambda^2 S = 2\lambda^2 \frac{\partial W}{\partial C} = \lambda \frac{dW}{d\lambda}.$$

Hyperelasticity: Stress from a Stored-Energy Function

A **hyperelastic** material stores energy W (deformation is path-independent).

3D energy $W(\mathbf{C})$, stress $\mathbf{S} = 2 \partial W / \partial \mathbf{C}$

$$W_{\text{neo-Hooke}} = \frac{c}{2} (I_1 - 3), \quad W_{\text{Fung}} = \frac{c_1}{4c_2} \left[e^{c_2(I_1-1)^2} - 1 \right]$$

Invariants of $\mathbf{C}$: $I_1 = \text{tr } \mathbf{C}$ (isotropic — elastin); $I_4 = \mathbf{M} \cdot \mathbf{C} \mathbf{M} = \lambda_{\text{fib}}^2$, the squared stretch along the reference fiber direction $\mathbf{M}$ (anisotropy — collagen).

0D reduction used here — one stretch λ , scalar $\sigma = \lambda \, dW/d\lambda$



(7)

Kinematic Growth

Multiplicative Decomposition: Growth vs. Elasticity

Split the deformation into a stress-free **growth** part F_g and a stress-carrying **elastic** part F_e .

3D a product of tensors

$$\mathbf{F} = \mathbf{F}_e \mathbf{F}_g, \quad \text{only } \mathbf{F}_e \text{ stores energy.}$$

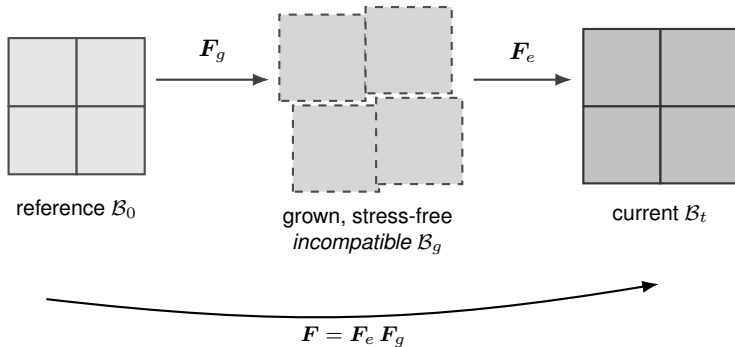
0D scalar stretches



(8)

Multiplicative Decomposition: Growth vs. Elasticity

F_g yields an *incompatible* stress-free state, F_e restores compability via residual stress.



Kinematic Growth Tensor

F_g encodes *how* the tissue is assumed to add mass. A common choice grows the tissue an amount θ along a direction f :

3D prescribed growth tensor

$$F_g = I + \theta f f^T \quad (9)$$

0D growth stretch

$$\lambda_g \text{ (scalar growth stretch),} \quad \theta = \det F_g = M/M_0.$$

Growth Law

The growth *pattern* F_g is fixed; the growth law evolves its single scalar to null the stress deviation from the set-point (first-order, production $\propto$ current mass). Since F_g is fixed by θ , its rate $\dot{F}_g = \dot{\theta} \mathbf{f} \otimes \mathbf{f}$ reduces to *one* scalar ODE — the *same* law in the full 3D tensor model and the 0D reduction (both track the scalar θ):

(10)

The set-point $\bar{\sigma}_h$ is **prescribed**: wherever growth stops ($\dot{\theta} = 0$), stress is homeostatic *by construction*. So the model can only *impose* its set-point, never *predict* stability — that is what the mixture theories add.

⚙️ Setup: Example — hypertensive artery (ex02)

Example. A pressurised artery is made *hypertensive* by a sustained pressure step. Kinematic growth thickens the wall — does the wall stress $\bar{\sigma}/\bar{\sigma}_h$ return to the *prescribed* set-point?

Running the Exercises

One runner drives every exercise — no Python to edit. Run the input file: `uv run python run.py configs/ex02_kinematic_growth.yaml`

Results: the key numbers ($\bar{\sigma}/\bar{\sigma}_h$, mass, diverged) print to the terminal, and the time history opens as a plot.

Parametric study — one curve per value, from the command line: `... -sweep insult.pressure_factor=1.5,2,3`

Every parameter in the file can be swept or overridden the same way.

Your turn: Kinematic growth

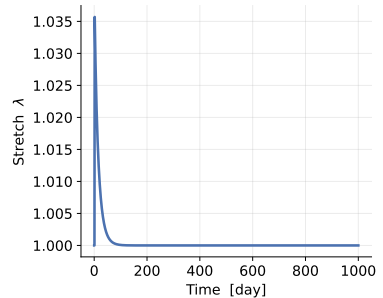
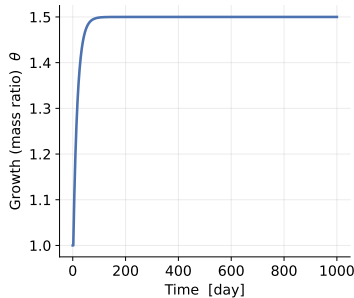
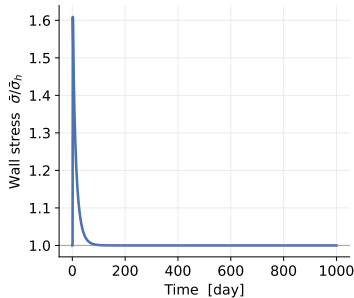
Question. Does the wall stress return to the prescribed set-point? Try larger pressure steps — does the outcome ever change?

Hints. $\bar{\sigma}/\bar{\sigma}_h$ settles at 1 while θ climbs to the pressure factor. Under pure hypertension it *always* adapts — runaway needs the mixture / aneurysm story (later).

✓ Answer: Kinematic growth

The stress plateau *always* lands on the prescribed set-point — the wall simply thickens more for a larger pressure step. Kinematic growth *imposes* stability; it cannot predict runaway.

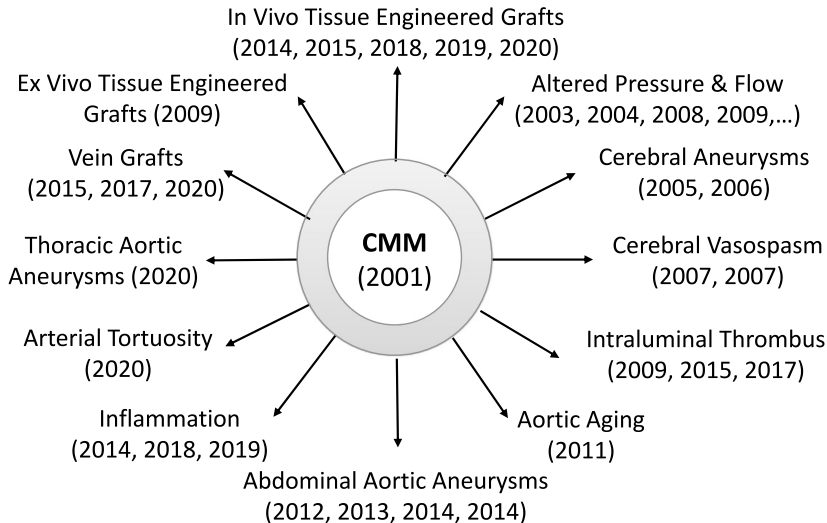
Kinematic growth of the artery under hypertension (P: 1.0 $\rightarrow$ 1.5x)



Learning: reaches the a-priori prescribed homeostasis — stability is built in, not predicted.

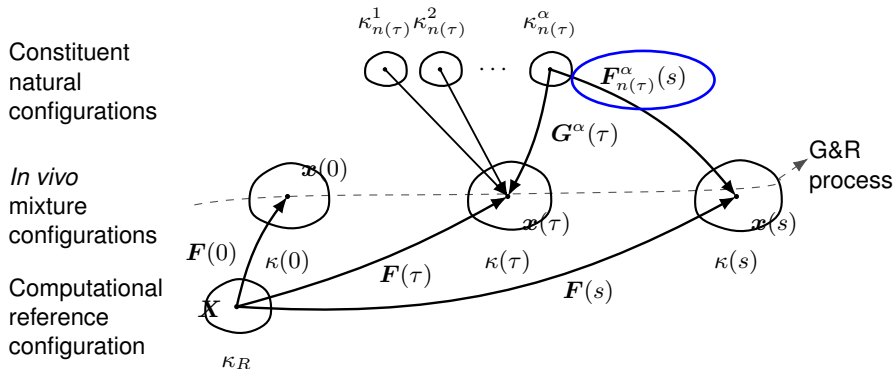
The Constrained Mixture Model

One Framework, Many Applications



Ubiquitous Vascular G&R

The Idea: Track Every Cohort



κ : configuration • τ : deposition time • s : current time • α : constituent • G^α : deposition stretch • $F_{n(\tau)}^\alpha$: constituent deformation

Load-Bearing Vessel Constituents

The wall is a mixture of three constituents, each with a reference **mass fraction** ϕ_0^α ($\sum_\alpha \phi_0^\alpha = 1$) and a **deposition stretch** G^α — the prestretch cells lay it down at, which *defines* its homeostatic stress $\sigma_h^\alpha = \sigma^\alpha(G^\alpha)$; the mixture set-point is $\bar{\sigma}_h = \sum_\alpha \phi_0^\alpha \sigma_h^\alpha$.

Constituent	Role	ϕ_0	G^α	law; turnover $1/k_d$
Elastin	bears load at low stretch	0.30	1.40	neo-Hooke; none (decades)
Collagen	stiff, crimped; anti-rupture	0.35	1.08	Fung; fast (~ 20 d)
Smooth muscle	passive + active; the sensor	0.35	1.20	Fung; fast (~ 20 d)

Elastin cannot be renewed ($k_d^e = 0$) — its loss drives *aneurysm*; collagen & SMC **turn over**, so the wall can *adapt*.

Every Cohort Shares One Stretch

A cohort of α deposited at τ is laid down at G^α and then stretches *with the mixture*: every constituent and cohort is **constrained** to the same tissue stretch $\lambda(s)$.

3D constituent-specific deformation

$$\mathbf{F}_{n(\tau)}^\alpha(s) = \mathbf{F}(s) \mathbf{F}^{-1}(\tau) \mathbf{G}^\alpha(\tau).$$

0D scalar elastic stretch

(11)

The shared $\lambda(s)$ is the *constraint*. At birth ($t = \tau$) the cohort sits at G^α (hence σ_h^α); older cohorts measure the same λ from an earlier $\lambda(\tau)$.

Mass Balance

Two G&R times: the *current* time s and a past *deposition* time $\tau \in [0, s]$. Track each constituent's **apparent mass density** ρ^α (mass of α per current mixture volume; $\rho = \sum_\alpha \rho^\alpha$).

Spatial mass balance, one per constituent:

$$\frac{\partial \rho^\alpha}{\partial s} + \underbrace{\operatorname{div}(\rho^\alpha \mathbf{v}^\alpha)}_{\text{transport}} = \underbrace{\bar{m}^\alpha}_{\text{net mass exchange}}, \quad \alpha = 1, \dots, N.$$

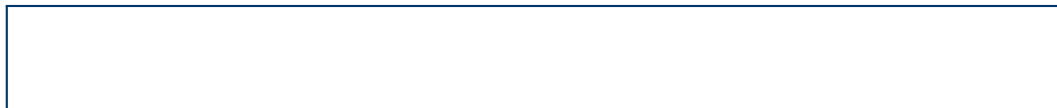
The net rate is *signed* — $\bar{m}^\alpha = \underbrace{m^\alpha}_{\text{production} \geq 0} - \underbrace{r^\alpha}_{\text{removal} \geq 0}$: < 0 atrophy • $= 0$ maintenance • > 0 growth.

From Balance to a Heredity Integral

Assumption 1 — constrained mixture: every constituent moves with the mixture, $v^\alpha = v$; as G&R is slow vs. the cardiac cycle the transport term is negligible, so $\partial \rho^\alpha / \partial s = \bar{m}^\alpha$ integrates directly in time.

Assumption 2 — prescribed rates: a true production rate $m^\alpha(\tau) \geq 0$ and a survival fraction $q^\alpha(s, \tau) \in [0, 1]$. Integrating from the distant past to s :

(12)



with $Q^\alpha(0) = 1$. **Assumption 3:** bulk density is nearly constant, $\rho(s) = \rho(0)$ — mass changes track volume changes.

Linear Momentum Balance

A rule-of-mixtures for the stresses (no body forces) satisfies the classical linear-momentum balance:

$$\operatorname{div} \boldsymbol{\sigma} = \rho \boldsymbol{a} \xrightarrow{\text{quasi-static}} \operatorname{div} \boldsymbol{\sigma} = \mathbf{0}.$$

Quasi-static assumption: G&R evolves slowly at every time s , so the inertia $\rho \boldsymbol{a}$ is negligible — the tissue stays in mechanical equilibrium throughout. G&R is then a *sequence* of finite-elasticity boundary-value problems.

Constitutive Relations: Cauchy Stress

In finite elasticity the Cauchy stress follows from a stored-energy function W (per unit reference mixture volume):

$$\boldsymbol{\sigma}(s) = \frac{2}{J} \mathbf{F}(s) \frac{\partial W(s)}{\partial \mathbf{C}(s)} \mathbf{F}^T(s), \quad \mathbf{C} = \mathbf{F}^T \mathbf{F}, \quad J = \det \mathbf{F}.$$

Rule of mixtures: $W = \sum_{\alpha} \tilde{W}^{\alpha}$. Closing the model needs **three constituent-specific relations** — a *stored energy* W^{α} , a *production rate* m^{α} , and a *removal* (survival) q^{α} .

Three Constituent-Specific Constitutive Relations

1. Material — constituent stored energy W^α : neo-Hookean elastin, Fung-type collagen & smooth muscle (eq. (7)).

2. Rate of production — up with wall stress & inflammation, down with wall shear:

(13)



$\Delta\sigma$: pressure • $\Delta\tau_w$: flow • $\Delta\rho_\varphi$: inflammation

3. Rate of removal — a survival fraction that decays faster under stress perturbation:

(14)



Constituent Stored-Energy Functions

Each cohort carries its *own* stored energy W^α , evaluated at its elastic deformation

$\mathbf{F}_{n(\tau)}^\alpha(s) = \mathbf{F}(s) \mathbf{F}^{-1}(\tau) \mathbf{G}^\alpha(\tau)$. Summing over cohorts with the *same* heredity structure gives the apparent (mixture-level) energy:

(15)



The mixture energy $W = \sum_\alpha \tilde{W}^\alpha$ closes the stress above.

Special Cases: $s = 0$ and Homeostasis

(i) At $s = 0$ (just before a perturbation) only original material is present, and $Q^\alpha(0) = 1$:

$$\rho \tilde{W}^\alpha(0) = \rho^\alpha(0) W^\alpha(\underbrace{\mathbf{F}_{n(0)}^\alpha(0)}_{\phi^\alpha(0)}) \Rightarrow \tilde{W}^\alpha(0) = \underbrace{\frac{\rho^\alpha(0)}{\rho(0)}}_{\phi^\alpha(0)} W^\alpha(0),$$

recovering the rule of mixtures $W(0) = \sum_\alpha \phi^\alpha(0) W^\alpha(0)$.

(ii) **Homeostasis** — turnover in an *unchanging* configuration, so $\mathbf{F}_{n(\tau)}^\alpha(s) = \mathbf{F}_{n(0)}^\alpha(s)$ (independent of τ). Then W^α factors out and, using the mass balance,

(16)



Perfect Maintenance

Take *constant* rates: $m^\alpha = m_0$, $Q^\alpha(s) = e^{-K^\alpha s}$, $q^\alpha(s, \tau) = e^{-K^\alpha(s-\tau)}$. The mass balance integrates to

$$\rho^\alpha(s) = \underbrace{\rho^\alpha(0) e^{-K^\alpha s}}_{\text{original decays}} + \underbrace{\frac{m_0}{K^\alpha} (1 - e^{-K^\alpha s})}_{\text{production accumulates}}.$$

A *time-invariant* density ($\rho^\alpha(s) = \rho^\alpha(0)$ for all s) requires the two to balance exactly:

(17)



Production m_0 then exactly offsets removal $K^\alpha \rho^\alpha$ — continuous turnover, unchanging tissue.

Your turn: Computational Cost?

Question. At each of N time steps the mixture stress sums over *all* cohorts deposited so far. How does the total cost scale with N ?

Hints. Step k re-sums $\sim k$ stored cohorts; add up $\sum_{k=1}^N k$.

$$\sum_{k=1}^N k = \frac{N(N+1)}{2} = \mathcal{O}(N^2)$$

⚙️ Setup: Example — turnover under hypertension (ex03)

Example. The same artery, now with collagen & SMC turning over (elastin does not). Which constituent grows to carry the load, and does turnover restore the set-point? Equilibrium each step: $\bar{\sigma}(\lambda) = \bar{\sigma}_h \gamma \lambda^2 / m$ (load factor γ , mass ratio m).

Run it. `uv run python run.py configs/ex03_constrained_mixture.yaml`

Sets each constituent's `turnover_days` ($1/k_d$), `gain` (K_σ), and the `insult` — hypertension (γ) or aneurysm (`elastin_surviving`).

Parametric study. ... `-set model.constituents.collagen.gain=3.0`

Your turn: Constrained mixture

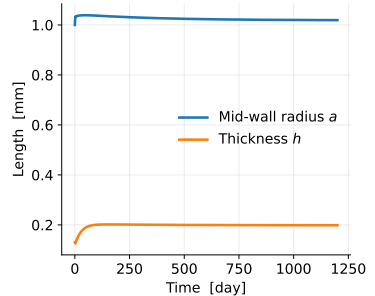
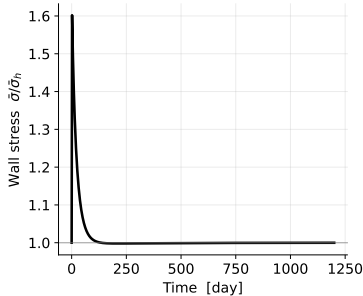
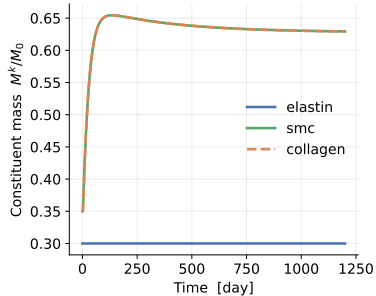
Question. Does the same tissue *adapt* to hypertension but *run away* on the aneurysm insult? Which constituent grows to carry the load elastin used to bear?

Hints. Try turnover 10 / 20 / 70 d and gain 0.3 / 1.0 / 3.0. Faster turnover and higher gain reach the new state sooner; watch which insult the tissue can absorb.

✓ Answer: Constrained mixture

Mild hypertension **adapts**: collagen and SMC grow, tissue stress returns to homeostatic. A severe elastin insult can **run away**. Faster turnover / higher gain reach the new state sooner.

Full constrained-mixture model: adaptation to hypertension (artery)



Homogenized CMM

Homogenize: One Mean Natural Configuration per Constituent

The full model tracks *every* past cohort with its own natural configuration ($O(N^2)$). **Temporal homogenization** instead gives each constituent a *single, evolving* mean natural configuration — its whole mass shares one homogenized elastic deformation, so two ODEs per constituent replace the entire history ($O(N)$).

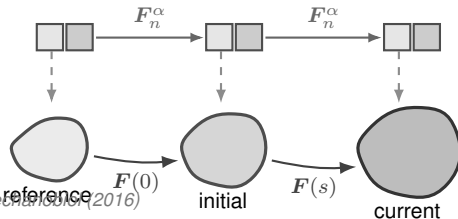
3D general form (Cyron 2016) — homogenized elastic deformation

$$\mathbf{F}_e^\alpha(s) = \mathbf{F}(s) [\mathbf{F}_n^\alpha(s)]^{-1}$$

0D 0D reduction — scalar mean natural stretch

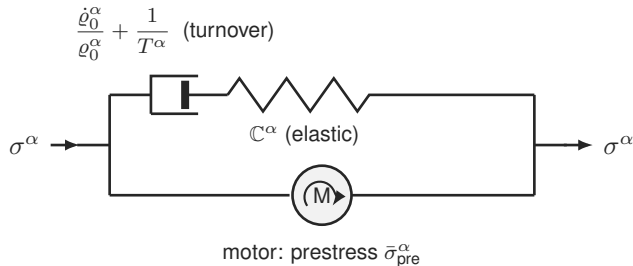
$$\lambda_e^\alpha = \lambda / \lambda_n^\alpha$$

stress-free *mean* natural configuration per constituent (*evolving* — not a history)



The Mechanical Analog: A Maxwell Fluid with a Motor

Each constituent behaves like a **Maxwell element in parallel with a motor**: turnover relaxes stress toward a *prestress*, not zero.



σ^α	constituent stress (the load)
$\mathbb{C}^\alpha$	elastic stiffness (spring)
$\dot{\varrho}_0^\alpha / \varrho_0^\alpha + 1/T^\alpha$	turnover + degradation (dashpot)
$\bar{\sigma}_{\text{pre}}^\alpha$	deposition prestress (motor)

Back to the full CMM (0D):

- spring $\leftrightarrow$ constituent law $\sigma^\alpha(\lambda_e^\alpha)$
- dashpot $\leftrightarrow$ mass turnover / removal k_d^α
- motor $\leftrightarrow$ deposition stretch G^α

Growth (Mass) and Remodeling (Natural Configuration)

Two 0D ODEs per constituent replace the cohort history. **Growth**: mass production $\propto$ mass, driven by the tissue-stress deviation. **Remodeling**: new mass enters at natural stretch λ/G^α , old mass leaves at the current λ_n^α ; mixing the two at the turnover rate evolves the mean.

0D growth — mass



(18)

0D remodeling — mean natural stretch



(19)

with production $m^\alpha = k_d^\alpha M^\alpha \Upsilon^\alpha$. Holding λ fixed, remodeling relaxes each constituent's stress to its prestress σ_h^α (time constant $\sim 1/k_d$) — the Maxwell-with-motor behaviour.

✂ Your turn: Quick Check — Why Is the Homogenized Model Cheap?

Question. Its entire state is $(M^\alpha, \lambda_n^\alpha)$ per constituent — a *fixed* number of values, updated once per step. How does its cost scale with the number of steps N , compared with the full model's $O(N^2)$?

Hints. No growing history to re-sum: constant work per step.

homogenized: $O(N)$ vs. full CMM: $O(N^2)$

Same time trajectory, far cheaper — and the fixed state maps onto a viscoelastic (Maxwell-with-motor) material, so it drops into standard finite-element solvers.

✂ Your turn: When do homogenized and full agree?

Question. The homogenized model keeps only the *mean* natural configuration per constituent. Under what assumptions should it match the full deposition-history model — and when would you expect them to diverge?

Resources. The two remodeling pictures: full heredity integral vs. the single mean stretch λ_n^α (19).

Hints. Think about the *spread* of natural configurations across cohorts. When is a single mean a good summary of the whole history?

✓ Answer: When do homogenized and full agree?

They agree when each constituent's cohorts share *nearly the same* natural configuration — i.e. slow, gradual loading where deposition stretch G^α and turnover are roughly stationary, so the cohort spread stays narrow and the mean represents it well. **They diverge** under fast or strongly time-varying insults, when cohorts deposited at very different states coexist and a single mean stretch can no longer capture the distribution.

⚙️ Setup: Example — homogenized vs. full (ex04)

Example. Run both mixture theories on the *same* artery and insult. The homogenized model keeps one mean natural stretch per constituent (two ODEs, (18)–(19)) instead of the full cohort history. How closely does it track the full model, and how much faster is it?

Run it. `uv run python run.py configs/ex04_homogenized_vs_full.yaml`

Here study: compare runs theories: [constrained_mixture, homogenized_cmm] on one scenario and prints each runtime and the largest mass-ratio gap between them.

Parametric study. Stress the comparison with a harsher, longer insult: `... -set insult.elastin_surviving=0.3 -set insult.ramp=10`
`... -set simulate.t_end=4000 # does the agreement hold?`

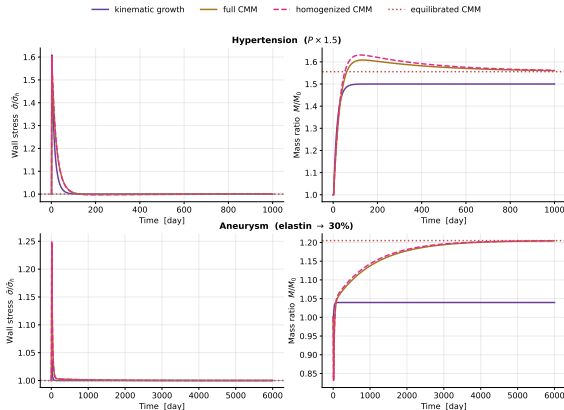
Your turn: Homogenized vs. full

Question. Is the difference between the two curves visible by eye? By what factor is the homogenized model faster — and does the agreement survive a harsher, longer insult?

Hints. Read the printed mass-ratio gap and the two runtimes. The homogenized model should track the full one closely wherever the loading is slow and gradual.

✓ Answer: Homogenized vs. full

The homogenized curve overlays the full CMM closely at a fraction of the runtime ($O(N)$ vs. $O(N^2)$). **Learning:** it reproduces a **similar time trajectory** to the full model — cheaply.



Equilibrated CMM

Skip the Transient — Solve for the End State

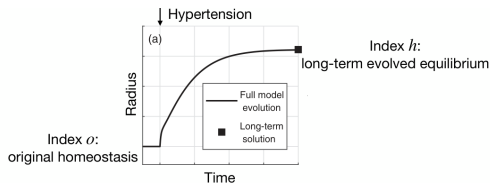
Set every rate to zero and solve the **algebraic** equilibrium — one root-find instead of thousands of time steps.

3D general form — the 3D equilibrium conditions

$$\underbrace{\Upsilon = 1 \Rightarrow \bar{\sigma}^* = \bar{\sigma}_h}_{\text{(A) growth balance}} \quad \underbrace{F_n^\alpha = \text{const} \Rightarrow \sigma^\alpha = \sigma_h^\alpha}_{\text{(B) remodeling done}} \quad \underbrace{\nabla \cdot \sigma = 0}_{\text{(C) mechanical equilib.}}$$

0D reduction — scalar conditions at the evolved geometry

$$\bar{\sigma}^* = \bar{\sigma}_h, \quad \lambda_e^\alpha = G^\alpha, \quad \sigma_\theta = \frac{Pa}{h}.$$



Collapse to One Scalar Equation

Let the turnover constituents grow in fixed proportion (exact when they share gain and turnover). The 3D conditions (A)–(C) collapse to a *single* equation for the evolved stretch λ^* — consistent with the same λ , G^α , ϕ^α used throughout:

0D scalar equilibrium equation for λ^*

(20)

with the Laplace exponent $n = 2$, γ the sustained load factor, s the surviving elastin fraction, ϕ^e the elastin mass fraction. Once λ^* is known, the evolved mass ratio is $M/M_0 = \gamma(\lambda^*)^2$.

This equation *is* the fixed point of the homogenized ODEs ((18)–(19) with $d/dt = 0$), so its root **coincides with the long-time limit of the transient theories** — to several digits.

The Punchline: Existence = Stability

Equation (20) may have **no physical root** — then no adapted state exists and the transient theories dilate without bound. The solver is therefore the cleanest **stability test**: root $\Rightarrow$ adapts, no root $\Rightarrow$ unstable. Near the boundary the transient slows critically, which is exactly why the instant algebraic test is so useful.

⚙️ Setup: Example — the equilibrated stability test (ex05)

Example. Skip the transient: solve the single algebraic equation (20) for the *final* adapted stretch λ^* . A physical root means the tissue adapts; *no root* means it dilates without bound. Scanning elastin loss then traces the stability boundary.

Run it. `uv run python run.py configs/ex05_equilibrated.yaml`

Here `study: equilibrium` solves once for the given insult, then the `scan: block` sweeps `elastin_surviving` and `report_critical: elastin` prints the critical surviving fraction s_{crit} .

Parametric study. Move the boundary by stiffening the wall, then re-read s_{crit} : ... `-set model.constituents.collagen.gain=3.0`
... `-set model.constituents.collagen.law.c1=600`

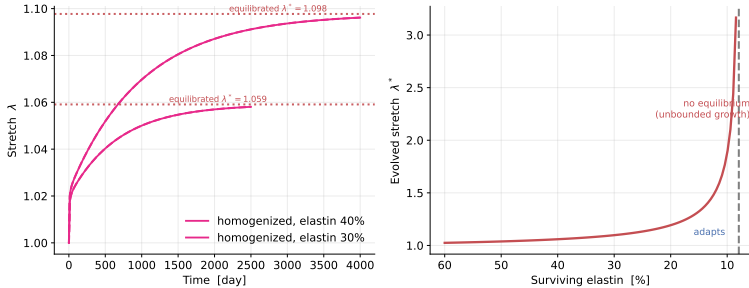
Your turn: Equilibrated

Question. Does the algebraic λ^* land on the transient plateau where a root exists — and what is the critical elastin loss beyond which *no* equilibrium exists at all?

Hints. The scan curve shoots up as $s \rightarrow s_{\text{crit}}$: the adapted stretch diverges just before the root vanishes. Below s_{crit} the transient models dilate without bound.

✓ Answer: Equilibrated

Where a root exists, the equilibrated solve lands exactly on the transient end-state. Below s_{crit} there is *no* root — the transient dilates without bound.



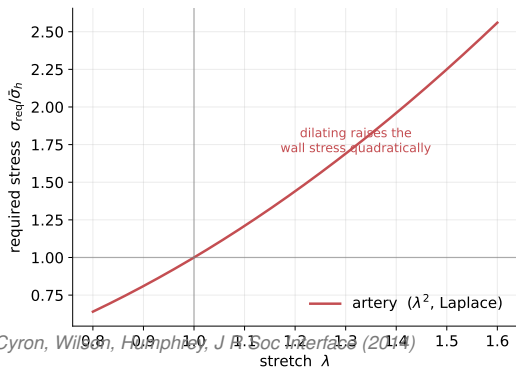
Learning: **matches** the transient theories *if* an equilibrium exists — and flags when it does not.

Stability

Where Instability Comes From: The Laplace Exponent

In a pressurised tube the wall stress is $\bar{\sigma} = \frac{Pa}{h}$ with radius $a \propto \lambda$ and thickness $h \propto M$, so the required (Laplace) stress is *quadratic* in the stretch, $\bar{\sigma}_{\text{req}} \propto \gamma \lambda^2 / M$. The homeostatic state is stable only if adding mass *lowers* the stress it must carry:

(21)



Growth thickens the wall ($h \propto M$), which lowers stress; but dilation raises it as λ^2 (Laplace).

While enough **elastin** survives, its stiff recoil keeps $d\bar{\sigma}/d\theta < 0$ and the artery **adapts**. Lose enough elastin and the sign flips, $d\bar{\sigma}/d\theta > 0$ — **runaway**. It is the quadratic Laplace exponent that makes this possible.

Same Tissue, Two Insults

Read stability two ways: **existence** — the equilibrated equation (20) has a root iff the tissue can adapt (no time integration); or **transient** — integrate and watch it plateau or climb. The same tissue can do either: mild elastin loss adapts, severe loss dilates without bound.

⚙️ Setup: Example — the stable/unstable map (ex06)

Example. A tissue *adapts* to an insult iff a mechanobiological equilibrium exists (20). Sweeping that instant test over (surviving elastin) $\times$ (pressure) draws the stable/unstable map. Then change *one* tissue property and watch the unstable “aneurysm” region shrink.

Run it. `uv run python run.py configs/ex06_stability.yaml`

Here study: map builds the 2D grid over the two axes: via the adapts existence test and shades adapt (blue) vs. runaway (red); it also prints the fraction of the grid that adapts.

Parametric study. Change one property in the `model`: block (gain, collagen `c1`, or G) — e.g.

```
... -set model.constituents.collagen.gain=3.0
```

```
... -set model.constituents.collagen.G=1.15
```

— and compare how much the stable region grows.

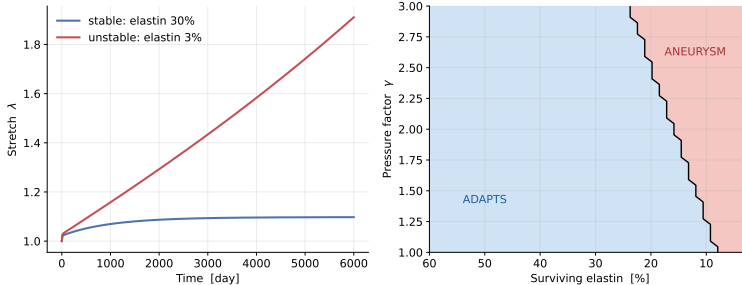
Your turn: Stability map

Question. Which single change — higher gain, stiffer collagen, or larger deposition stretch — enlarges the stable region the most? Elastin cannot be regrown in adults: what does that imply for aneurysm?

Hints. Compare the printed “fraction that adapts” across the three changes. Each shifts the boundary between adaptation and runaway.

✓ Answer: Stability map

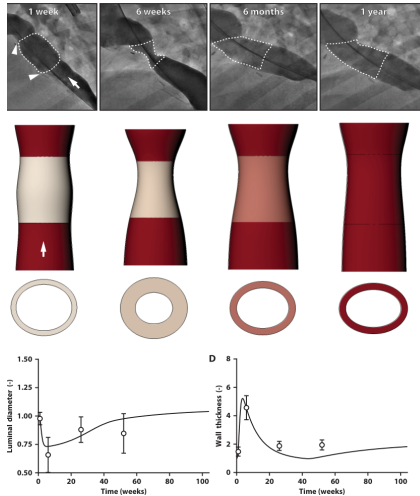
Higher gain, stiffer collagen, or larger deposition stretch all enlarge the *adapts* region — the tissue tolerates a stronger insult before running away.



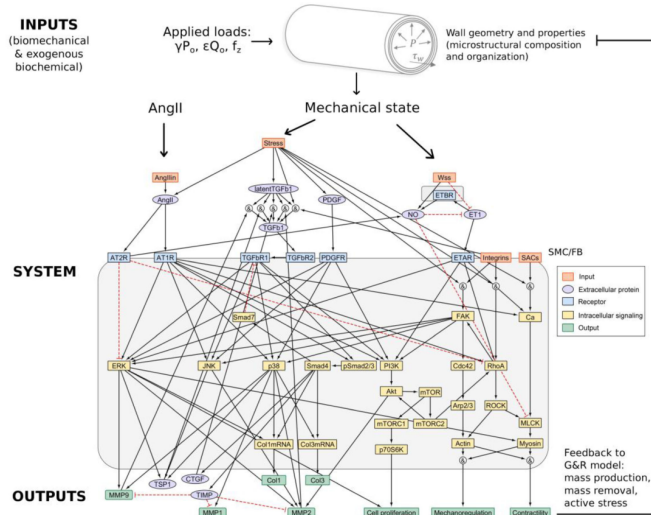
Learning: the existence test that defines the equilibrium boundary *is* the stability criterion.

Future Perspectives

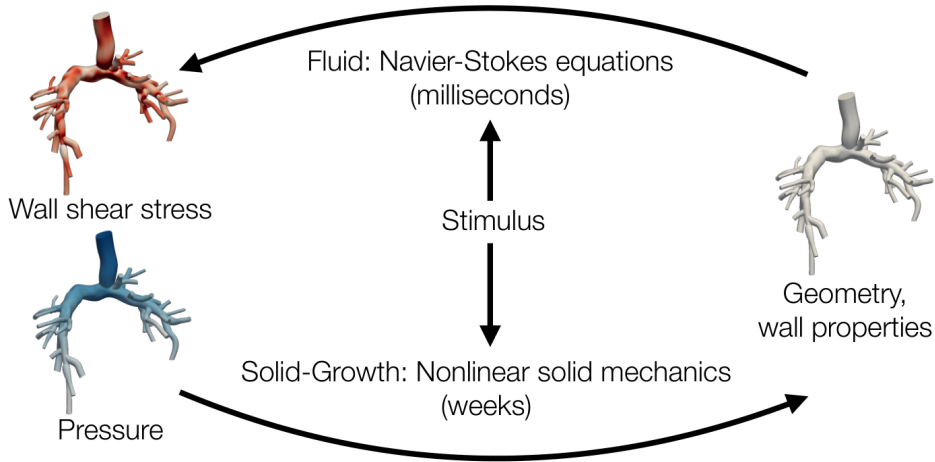
Clinical Application: Reversible Graft Stenosis



Systems-Biology Coupling: Signaling to Remodeling



Fluid–Solid–Growth: Closing the Hemodynamic Loop

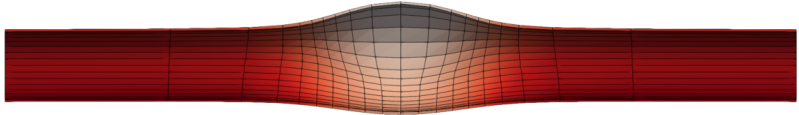


3

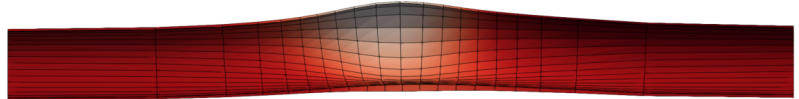
FSGe: Long-Term Evolved Aneurysmal Equilibrium

Solid with $\tau_w \sim 1/r^3$

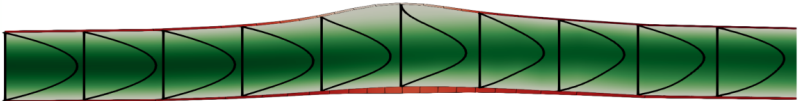
Wall shear
stress



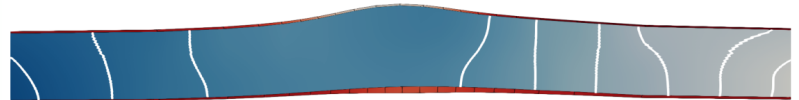
**Fluid-Solid-Growth
interaction**



Velocity



Pressure



Open-Source Implementations

Under the Hood — What Each File Computes (1/2: the core)

The `gr` package is layered: a shared **continuum-mechanics core** (read, don't edit), then one file per G&R theory.

File	Computes	Continuum mechanics / G&R
<code>mechanics.py</code>	strain energy $W(\lambda_e)$, 2nd PK stress $S = dW/dE_e$, Cauchy $\sigma = \lambda_e^2 S$ (neo-Hookean, Fung)	finite-strain hyperelasticity of each constituent, in reference measures
<code>parameters.py</code>	Model/Constituent $(\phi_0, G^\alpha, k_d, K_\sigma)$; set-point $\sigma_h^\alpha = \sigma^\alpha(G^\alpha)$; Insult	deposition prestretch & homeostatic set-points ; the two insults
<code>geometry.py</code>	required Laplace stress $\bar{\sigma}_h \propto \lambda^2/m$; solves $\bar{\sigma}(\lambda) = \sigma_{\text{req}}$ for λ	mechanical equilibrium ; no root $\Rightarrow$ runaway
<code>history.py</code>	Result: $t, \lambda, M/M_0, \bar{\sigma}, \dots$	shared time-history output

Under the Hood — What Each File Computes (2/2: the theories)

File	Computes	G&R theory / role
kinematic_growth.py	split $\mathbf{F} = \mathbf{F}_e \mathbf{F}_g$, scalar θ ; $\dot{\theta} = k_g \theta (\bar{\sigma} / \bar{\sigma}_h - 1)$	Theory 1 — kinematic growth
constrained_mixture.py	heredity integrals over cohorts; mass balance + mixture stress ($O(N^2)$)	Theory 2 — full CMM
homogenized_cmm.py	two ODEs per constituent: $\dot{M}^\alpha, \dot{\lambda}_n^\alpha$ ($O(N)$)	Theory 3 — homogenized CMM
equilibrated_cmm.py	rates $\rightarrow 0$: one algebraic solve for λ^*	Theory 4 — equilibrated CMM
stability.py	adapts (does a root exist?), critical_elastin_loss	stability = existence of an equilibrium
scenario.py, configs/ run.py,	build model/geometry/insult from YAML; run a study; sweep	the exercise driver (input files)

Common thread: each theory changes only *how the supplied stress $\bar{\sigma}(\lambda)$ is built* — the same constituent laws and equilibrium solve are reused, so every difference you saw is the G&R law alone.

Research-Grade Open-Source Solvers

Project	What it does	Stack / group
svGrowth	constrained-mixture G&R of vessels (biaxial thin-wall); the cross-validation reference used above	Python; Stanford CBCL
svFSGe	fast, strongly-coupled 3D fluid–solid–growth driver (the FSGe method)	Python + svFSI/svMultiPhysics; Stanford CBCL
FEMBE_Plugin	mechanobiologically equilibrated constrained-mixture material as a FEBio plugin	C++ / FEBio; Yale (Humphrey lab)
FEFSG_Plugin	fluid–solid–growth coupling as a FEBio plugin (thoracic-aorta hypertension & aneurysm cases)	C++ / FEBio; Yale (Humphrey lab)

github.com/StanfordCBCL/svGrowth github.com/StanfordCBCL/svFSGe

github.com/yale-humphrey-lab/FEMBE_Plugin github.com/yale-humphrey-lab/FEFSG_Plugin

How the Four Theories Answer — The Takeaways

Theory	Stability?	What it says
Kinematic growth	<i>Imposed</i>	Either reaches the <i>a-priori prescribed</i> homeostasis, or is unstable — nothing in between. Built in, not learned.
Constrained mixture (full)	<i>Predicted</i>	Adapts or dilates depending on the insult — the faithful reference.
Homogenized CMM	<i>Predicted</i>	A similar time trajectory to the full model, at a fraction of the cost.
Equilibrated CMM	<i>Instantly</i>	Matches the transient theories <i>if</i> an equilibrium exists; <i>no</i> root $\Rightarrow$ it does not match — unbounded growth.

Summary — What to Take Away

The biology. Living tissue continuously **turns over**; cells deposit new material *pre-stretched* at G^α , chasing a homeostatic mechanical set-point σ_h .

The four theories, one set-point, one question — does it adapt or run away?

- **Kinematic growth** — prescribe F_g ; *imposes* the set-point. Outcome is binary: reach the prescribed homeostasis, or go unstable.
- **Constrained mixture** — track every cohort; stability is *predicted* and **depends on the insult** (mild $\rightarrow$ adapt, severe $\rightarrow$ runaway).
- **Homogenized** — one mean natural configuration per constituent; a **similar time trajectory** to the full model at $O(N)$ not $O(N^2)$.
- **Equilibrated** — solve for the end state directly; **matches** the transient theories *iff* an equilibrium exists, and flags instability by finding no root.

Where it goes. These tools now drive **clinical prediction** (TEVGs), **systems-biology coupling**, and **fluid–solid–growth**.

Key references I

- Rodriguez, Hoger, McCulloch. Stress-dependent finite growth in soft elastic tissues. *J Biomech* (1994).
- Humphrey, Rajagopal. A constrained mixture model for growth and remodeling of soft tissues. *Math Models Methods Appl Sci* (2002).
- Cyron, Wilson, Humphrey. Mechanobiological stability. *J R Soc Interface* (2014).
- Cyron, Aydin, Humphrey. A homogenized constrained mixture (and mechanical analog) model. *Biomech Model Mechanobiol* (2016).
- Latorre, Humphrey. A mechanobiologically equilibrated constrained mixture model. *ZAMM* (2018).
- Latorre, Humphrey. Fast, rate-independent finite-element implementation of a 3D constrained mixture model. *Comput Methods Appl Mech Eng* (2020).
- Humphrey. Constrained mixture models of soft tissue growth and remodeling. *J Elasticity* (2021).
- Irons, Humphrey. Cell signaling model for arterial mechanobiology. *PLoS Comput Biol* (2020).
- Irons, Latorre, Humphrey. From transcript to tissue: multiscale modeling. *Ann Biomed Eng* (2021).
- Drews, Pepper, Best, . . . , Humphrey, Breuer. Spontaneous reversal of stenosis in tissue-engineered vascular grafts. *Sci Transl Med* (2020).

Key references II

- Figueroa, Baek, Taylor, Humphrey. A computational framework for fluid–solid–growth modeling. *Comput Methods Appl Mech Eng* (2009).
- Schwarz, Pfaller, . . . , Marsden. A fluid–solid–growth solver for cardiovascular modeling. *Comput Methods Appl Mech Eng* (2023).
- Pfaller, Latorre, . . . , Marsden. FSGe: a fast, strongly coupled 3D fluid–solid–growth method. *Comput Methods Appl Mech Eng* (2024).
- Humphrey, Eberth, Dye, Gleason. Fundamental role of axial stress in arterial mechanics. *J Biomech* (2009).
- Ambrosi, Ben Amar, Cyron, . . . , Humphrey, Kuhl. Growth and remodelling of living tissues. *J R Soc Interface* (2019).

Key references III

Code

- Seminar code (this deck's figures): `github.com/YaleCBL-Teaching/Growth-Remodeling`
- `github.com/StanfordCBCL/svGrowth`, `svFSGe`
- `github.com/yale-humphrey-lab/FEMBE_Plugin`, `FEFSG_Plugin`

Figure sources. Result figures reproduced with the `gr` package; literature figures cited per-slide.